INDEX

Acceleration vector, 15–18
Affective future, 100, 101
Affective past, 100
Angular momentum, conservation of, 29
Angular orientation, and physical laws, 6–8
Antiatoms, 44
Antimatter, 43–46
Antineutrons, 44
Antiparticles, 44
Antiprotons ($\bar{p}$), 44
Atomic bomb, energy of, 89–90
Atoms
  anti-atoms of, 44
  conservation of energy and, 134–136
  mass at rest, 89–90
  reversibility in time and, 29
  scale of, 27
Axial vectors, 35–38

Beta decay, 40–43, 45
Billiard bills, collision of, 51–52

Centrifugal force, 76
Challenger (space shuttle), xix, xvi
Clocks
  slowing in moving system, 59–62
  speed in gravitational field, 131–136, 140–141
Cobalt, 41–42, 45
Collision, analysis of, 83–88, 105
  inelastic, 87–88, 89
Conservation laws
  energy, conservation of. *See* Energy, conservation of
  four-vectors, 106–107
  mass, conservation of, 88, 89
  momentum, conservation of. *See* Momentum, conservation of
  parity, conservation of, 40–43
  symmetry and, 29–30
Conspiracy as law of nature, complete, 58
Current time (space-time region), 99–101
Curved space (curvature), 111–144
  defined, 117, 120, 122, 123, 125
  Einstein's law for, 126–128
  geometry of, 117–125, 128–129